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Refog, John the Ripper, or Prey Assignment

Cyber Risk Management

CIDM 6341

Question 1: what did you do? Provide a description of what you did. In particular, I want to know what you did to research the subject of the report. In most cases the value of the research is in the journey more than the result, same here, the better/clearer/insightful your write up is the better it scores.

To start this assignment, I chose to download the Refog version of the keylogging software, which was Hoverwatch. I watched the video on how to download Hoverwatch, and then started the process. For me to be able to download Hoverwatch, I had to override the dangerous site warning that Google provided, as well as override the dangerous download settings manually. Once I did that, I was able to complete the download, which I stored within its own file on my computer titled "DANGERZONE." I then opened Hoverwatch on my desktop, which required I change certain permissions for it to be fully functional. These permissions included allowing the app to function in the background through the "App Background Activity" in settings, as well as allowing the screen recording to occur. Once these permissions were granted, the online Dashboard began to record and track what was occurring on my laptop. At first, I misunderstood what all the app was tracking. I thought that it would track all activities I performed on my laptop, including all websites visited, everything typed, etc. It was not until I went to my shoe site to do some "shopping" for the keylogger to track that I realized that while it was tracking the websites, it was more focused on tracking the typing occurring. The software was able to track my key typing relatively well. I tracked it across several different apps, such as iMessage, Chrome, Mail, Excel, and Quicken to name a few. I tested how it tracked the "delete" button by going to my Mail app and deleting several emails using the "delete" button on my keyboard. This was not registered as activity on the scan. The results are discussed in the next section. Once I gathered a sufficient dataset to review, I stopped the Hoverwatch scan and downloaded the results. I then selected the option to delete the device, which deleted all the log data with it, and then moved the files downloaded to the trash and permanently deleted them from my laptop. I then analysed the results from the scan.

Once I completed the Hoverwatch/keylogging section of this project, I started the John the Ripper/Password Cracker portion. I started this by going to the website for John the Ripper and reading the information provided. I then downloaded the version necessary for my Mac's operating system using the "brew install." Once that was done, I referred to the "how to use" page on the John the Ripper page, which stated that I needed to provide John with a password file for it to start trying to crack passwords from. I then used Gemini to help me troubleshoot the problems I was experiencing when attempting to run the John the Ripper

commands. The first command I attempted to run was the “zip2john” command, which works to prepare a password-protected ZIP file that the John the Ripper can then attempt to guess the password. When attempting to run this command, the response was “command not found.” I troubleshooted using Gemini, the results of which are in the next section.

The last part of this assignment was to download Prey, the tracking device. I did so by following the provided instructions and downloading the Prey software on the device I wanted to track. The chosen device was my iPhone. I downloaded the app from the App Store and then created an account through that. Once completed, I then was able to go to my laptop (the controlling device) and log in there to see the tracked device (my iPhone).

Question 2: what were the results?

Summarize and explain what your findings mean. Provide any printouts of products produced, this could be a report, a map, a table, etc.

The results for the Hoverwatch keylogging portion of this assignment varied. Hoverwatch is a mobile tracker used to “register all the key pressed and typed messages on the keyboard of the monitored Mac” (Hoverwatch, n.d.). A keylogger is very similar to what it sounds like, a tool that “records what a person types on a device” or logs the keys entered (CrowdStrike, 2023). A keylogger functions in one of two ways. The first way is through the hardware, which are physical devices that can be disguised as “computer cabling” or “in a USB adapter” used to track every keystroke (CrowdStrike, 2023). The second way is through software. Software keyloggers do not require a physical device, but rather is software downloaded “intentionally or inadvertently with malware” (CrowdStrike, 2023). There are three types of software keyloggers, form-grabbing keyloggers, JavaScript keyloggers, and API keyloggers. Form-grabbing keyloggers are usually “deployed on a website” and record data entered into a specific field, such as one that might prompt a user to enter their username or password (CrowdStrike, 2023). The JavaScript keylogger is a type that is also deployed through websites. It is “written in JavaScript” and runs scripts to “record every keystroke” that a user enters in a website (CrowdStrike, 2023). The last kind of keylogger is an API (application programming interface) keylogger, which uses an API running within an app to record keystrokes performed within the application (CrowdStrike, 2023). Hoverwatch most likely utilizes the API keylogger to perform the keylogging. This is not the only function of Hoverwatch, as it also can track phone calls, photos, locations, contacts, calendars, and many other things (Hoverwatch, n.d.). The results of running Hoverwatch in the background are shown below:



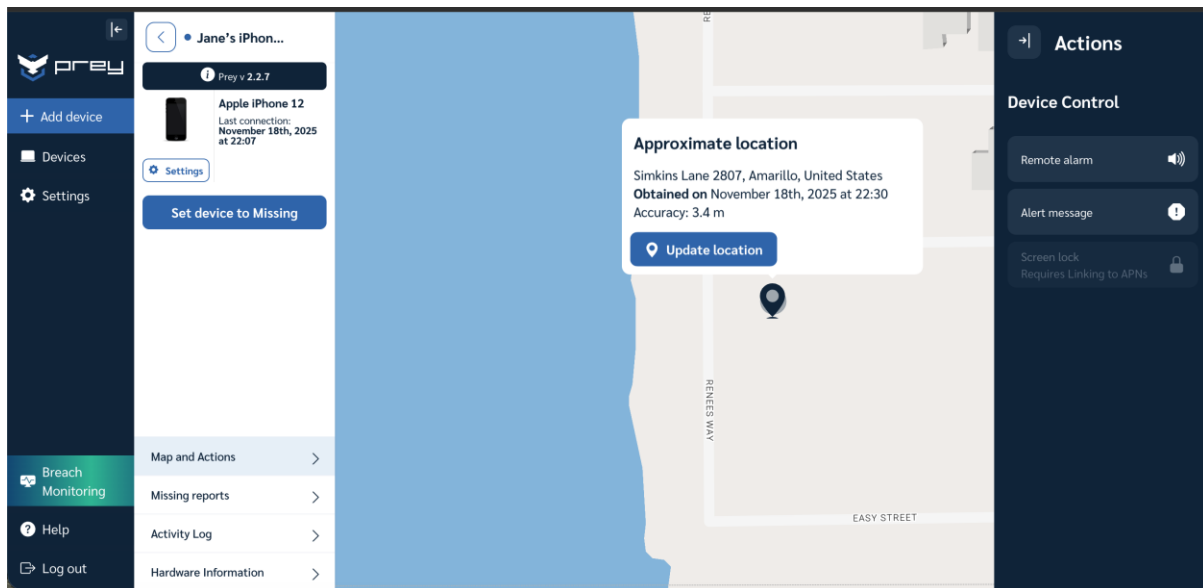
These results were presented in the format of “Date Created”, “Username”, “App”, “Executable”, and “Keystrokes typed.” I tested how well Hoverwatch tracked my activity but opening and using different apps and using different websites. Hoverwatch tracked my keystroke activity across the apps Google Chrome, Messages, Microsoft Excel, and Quicken (my banking application). The first thing I noticed was that Hoverwatch was able to track across many apps; however, for each of these apps it did not track much more than what was typed. For Google it did not track typing across different websites. There were several websites that I opened and was typing in, such as Google Gemini and Microsoft Docs, and while Hoverwatch tracked what I was typing, it did not track what websites I was typing what in. Furthermore, in Messages Hoverwatch was able to track what messages I was sending, but it did not track who those messages were sent to. When I was in Quicken, it only tracked that I typed “shootingshooting” which is one of the expense categories that I have in our Quicken. It did not track or show any additional information from Quicken from that.

John the Ripper is a password cracker program that is an important tool in good cybersecurity practices. John the Ripper can be an extremely useful and versatile tool for an organization to have to use for several reasons, as it can be used to recover lost or forgotten

passwords, as well as assess the strength of passwords during security audits. John the Ripper is very popular because of its ability to function across several different operating systems, as well as its ability to perform multiple types of attacks, including brute force, dictionary, and customized attacks. John the Ripper works by using a technique called hashing. A hash is a unique string of characters that represents a piece of data but is not the data itself. (edureka!, 2025). When I attempted to perform the password cracking using John the Ripper, the first command I attempted to run was the “zip2john” command. I continuously got the response “command not found.” Gemini said this issue was likely caused by the fact that the John the Ripper code was downloaded into a folder that was not included in the pathway, and as such the pathway was not correct and causing the problems with pulling the command. The file that contained the tools for John the Ripper were found in a support directory, which was ultimately the cause of the error, as Terminal did not automatically search the support directory for the command (Google, 2025). This issue was able to be resolved by including the full path to the command when performing the command. The second issue that I came across was that the required input file was not located in any of the expected directories on the system. This indicates that there is a file management issue that needs to be resolved.

```
janealdakkour ~ --bash -- 80x44
MacBook-Pro:~ janealdakkour$ brew unlink john-jumbo
Unlinking /opt/homebrew/Cellar/john-jumbo/1.9.0_1... 5 symlinks removed.
MacBook-Pro:~ janealdakkour$ brew install john
[=> Fetching downloads for: john
✓ Bottle Manifest john (1.9.0_1) [Downloaded 10.1KB/ 10.1KB]
✓ Bottle john (1.9.0_1) [Downloaded 13.7MB/ 13.7MB]
[=> Pouring john-1.9.0_1.arm64_tahoe.bottle.1.tar.gz
[=> /opt/homebrew/Cellar/john/1.9.0_1: 36 files, 20.6MB
[=> Running 'brew cleanup john' ...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
MacBook-Pro:~ janealdakkour$ zip2john
-bash: zip2john: command not found
MacBook-Pro:~ janealdakkour$ -bash zip2john
-bash: -bash: command not found
MacBook-Pro:~ janealdakkour$ zip2john
-bash: zip2john: command not found
MacBook-Pro:~ janealdakkour$ source ~/.bash_profile
MacBook-Pro:~ janealdakkour$ source ~/.zshrc
-bash: /Users/janealdakkour/.zshrc: No such file or directory
MacBook-Pro:~ janealdakkour$ zip2john
-bash: zip2john: command not found
MacBook-Pro:~ janealdakkour$ eval "$(/opt/homebrew/bin/brew shellenv)"
MacBook-Pro:~ janealdakkour$ echo $PATH
/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.13/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/Library/Apple/usr/bin:/Applications/nmap.app/Contents/Resources/bin:/Applications/ncat.app/Contents/Resources/bin:/Applications/nping.app/Contents/Resources/bin:/Applications/zenmap.app/Contents/Resources/bin
MacBook-Pro:~ janealdakkour$ zip2john
-bash: zip2john: command not found
MacBook-Pro:~ janealdakkour$ john passwd
stat: passwd: No such file or directory
MacBook-Pro:~ janealdakkour$ john --wordlist=password.lst --rules passwd
stat: passwd: No such file or directory
MacBook-Pro:~ janealdakkour$ # Install or reinstall John the Ripper
MacBook-Pro:~ janealdakkour$ MacBook-Pro:~ janealdakkour$ brew install john-the-ripper
-bash: MacBook-Pro:~: command not found
MacBook-Pro:~ janealdakkour$ # Convert a protected ZIP file into a crackable hash file named 'zip_hash.txt'
```

The results of the Prey device tracker are shown below:



The device was successfully and accurately tracked to its exact current location (the first request came from my house). The Prey Project is a company that offers the service to provide device security, management, and recovery to users. It works by installing an agent on the target device(s) such as computers or phones, which connects back to and regularly communicates with a central management web page control panel. Users are able to manage all the functionality of Prey from the management web page. Prey offers several functions for customers, which include tracking geo-location of devices, setting off security measures such as alarms or initiating a lock screen, and setting the device into “missing mode” which will send detailed reports back to the user (Prey Project, 2021). The detailed reports from “missing mode” include information such as location, network data, pictures from the device's camera, and screenshots. This information will be sent so long as the device is connected to the internet.

Question 3: what did you learn?

I can't tell you what you learned. What I will say is that I reward insight. Insights are aha moments (a term in use long before Oprah wanted to copyright it). If you see new ways of doing things, new insights to your thought processes, potential future applications be they personal or work related, crossovers to other topics, these are what I reward more than just telling me you learned lots. I expect you to learn lots, but it isn't till you explain where and what that I see that you really did. Ok, so sometimes you don't learn much. I'll still grade this area high if you tell me why, what you know, how this works on what you've done in the past, etc. Sometimes when you start doing this you see that what you've learned is reinforcing what you've done and sometimes you even have small aha moments. Bottom line is to be reflective, think a few minutes or overnight about what you've done and how it fits into your nomological net (your personal set of knowledge base structure, those theories and beliefs that guide how you evaluate and use knowledge). Then write the section, when I see this done, I always score the section higher.

While working on this assignment, I learned many things about how a bad actor can track information from a person's device using various hacking methods, such as keylogging, password cracking, and device tracking, as well as the kind of information that can be gathered from these methods. The first method I learned about was keylogging. I learned that keyloggers are tools that "records what a person types on a device" or logs the keys entered on a device (CrowdStrike, 2023). I learned about keylogging through using the program Hoverwatch. When I first started using Hoverwatch, I thought it would track all activity on the device, such as all website visited, all apps used, all things typed into the computer, all messages received, etc. I realised after letting it run for a couple of minutes that it was only tracking the activity when there was typing involved. I was also surprised to find that it did not register when the "delete" button was used in Mail. This is important to take note of as it means that the user could delete files, messages, emails, photos, or other things without it being tracked by Hoverwatch. This is possibly because of the version that I purchased as well, and there is likely a subscription that could be purchased that would allow the Hoverwatch admin to track the user's activity in more detail.

On the topic of password cracking, it was noted that John the Ripper is considered "advanced software," which I learned with performing this part of the project. I worked to troubleshoot several issues that I encountered while trying to utilize the software. My difficulties in using the John the Ripper software came in setting up the software with the prerequisite steps. This taught me the advanced technological knowledge one must have of the Homebrew support directory, the operating systems environmental variables, and the specific location of command files in the support directories. The theoretical knowledge I learned about how John the Ripper works behind this taught me about how it is useful and necessary in cybersecurity practices. I learned that John the Ripper can be used ethically by testing the strength of passwords on a network. This will help administrators eliminate weak passwords that put the network at risk. It can also be used to help users on a network set up strong passwords. Furthermore, should a password be forgotten, it allows for the opportunity to recover data from a locked device or folder through cracking the password. While it has many practical, ethical uses, there are still bad actors who will use it for bad.

Lastly, on the Prey Device Tracker topic, I learned about the Prey service that is offered. Prey is a geo-location management software that offers companies location and management services on network devices. This is a vital management and security measure for organizations, as it allows them to locate their devices and protect them in the event they are stolen. There are many different features that assist with this. The geolocation management system allows for an organization to track the near exact location of all devices registered and with the software downloaded. The other features that Prey has are alarm systems which can be triggered remotely from the control panel, send detailed reports (including location and camera photos of the location from the device), or to remotely initiate a lockdown of the device or wipe the device clean to prevent a bad actor from gaining access to vital information. This allows the network to protect itself from physical threats.

References:

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